7.2.3 Algebraic Operations on Matrices

Basic operations on matrices are

(1) multiplication of a matrix by a scalar,

(2) addition/subtraction of two matrices, and

(3) multiplication of two matrices.

There is no concept of dividing a matrix by another matrix and thus, the operation $\frac{A}{B}$, where A and B are matrices, is not defined.

(1) Multiplication of a matrix by a scalar

For a given matrix $A=[a_{ij}]_{m\times n}$ and a scalar k, we define a new matrix $kA=[b_{ij}]_{m\times n}$, where $b_{ij}=ka_{ij}$ for all i and j.

For instance, if $A=\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$, then $kA=\begin{bmatrix} ka & kb & kc \\ kd & ke & kf \end{bmatrix}$

In particular if $k=-1$, we obtain $-A=[-a_{ij}]_{m\times n}$. This $-A$ is called **negative** of the matrix A. Don't say $-A$ as a negative matrix.

(2) Addition and Subtraction of two matrices

If A and B are two matrices of the same order, then their sum denoted by $A + B$, is again a matrix of same order, obtained by adding the corresponding entries of A and B.

More precisely, if $A=[a_{ij}]_{m\times n}$ and $B=[b_{ij}]_{m\times n}$ are two matrices, then the sum $A + B$ of A and B is a matrix given by

$A+B=[c_{ij}]_{m\times n}$ where $c_{ij}=a_{ij}+b_{ij}$ for all i and j.

Similarly subtraction $A - B$ is defined as $A-B=A+(-1)B$.

That is, $A-B=[d_{ij}]_{m\times n}$, where $d_{ij}=a_{ij}-b_{ij}$ $\forall$ i and j. (The symbol $\forall$ denotes for every or for all).

Note 7.2

(i) If A and B are not of the same order, then $A + B$ and $A - B$ are not defined.

(ii) The addition and subtraction can be extended to any finite number of matrices.

Example 7.4

Compute $A + B$ and $A - B$ if

$$A=\begin{bmatrix} 4 & \sqrt{5} & 7 \\ -1 & 0 & 0.5 \end{bmatrix} \text{ and } B=\begin{bmatrix} \sqrt{3} & \sqrt{5} & 7.3 \\ 1 & \frac{1}{3} & \frac{1}{4} \end{bmatrix}.$$

Solution

By the definitions of addition and subtraction of matrices, we have

$$A+B=\begin{bmatrix} 4+\sqrt{3} & 2\sqrt{5} & 14.3 \\ 0 & \frac{1}{3} & \frac{3}{4} \end{bmatrix} \text{ and } A-B=\begin{bmatrix} 4-\sqrt{3} & 0 & -0.3 \\ -2 & -\frac{1}{3} & \frac{1}{4} \end{bmatrix}$$

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